

Exercise 1

Compute the first two iterations of the Jacobi method for solving the linear system $Ax = b$, with:

$$A = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solution

For computing $x^{(1)}$ and $x^{(2)}$, we can rely on the formula for the Jacobi method:

$$x_i^{(k)} = \frac{1}{a_{i,i}} \left(b_i - \sum_{j=1, j \neq i}^n a_{i,j} x_j^{(k-1)} \right)$$

Let us start from $x^{(1)} = [x_1^{(1)}, x_2^{(1)}, x_3^{(1)}]$:

$$x_1^{(1)} = \frac{1}{a_{1,1}} \left(b_1 - \sum_{j=2}^3 a_{1,j} x_j^{(0)} \right) = \frac{1}{4} (0 - 2 \cdot 0 - 0 \cdot 0) = 0$$

$$x_2^{(1)} = \frac{1}{a_{2,2}} \left(b_2 - \sum_{j=1, j \neq 2}^3 a_{2,j} x_j^{(0)} \right) = \frac{1}{2} (2 - 0 \cdot 0 - 1 \cdot 0) = 1$$

$$x_3^{(1)} = \frac{1}{a_{3,3}} \left(b_3 - \sum_{j=1}^2 a_{3,j} x_j^{(0)} \right) = \frac{1}{4} (0 - 1 \cdot 0 - (-1) \cdot 0) = 0$$

So $x^{(1)} = [0, 1, 0]$. Now we can compute $x^{(2)} = [x_1^{(2)}, x_2^{(2)}, x_3^{(2)}]$:

$$x_1^{(2)} = \frac{1}{a_{1,1}} \left(b_1 - \sum_{j=2}^3 a_{1,j} x_j^{(1)} \right) = \frac{1}{4} (0 - 2 \cdot 1 - 0 \cdot 0) = -\frac{1}{2}$$

$$x_2^{(2)} = \frac{1}{a_{2,2}} \left(b_2 - \sum_{j=1, j \neq 2}^3 a_{2,j} x_j^{(1)} \right) = \frac{1}{2} (2 - 0 \cdot 0 - 1 \cdot 0) = 1$$

$$x_3^{(2)} = \frac{1}{a_{3,3}} \left(b_3 - \sum_{j=1}^2 a_{3,j} x_j^{(1)} \right) = \frac{1}{4} (0 - 1 \cdot 0 - (-1) \cdot 1) = \frac{1}{4}$$

So $x^{(2)} = [-\frac{1}{2}, 1, \frac{1}{4}]$.

Exercise 2

Compute the first two iterations of the Gauss-Seidel method for solving the linear system $Ax = b$, with:

$$A = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & -1 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solution

For computing $x^{(1)}$ and $x^{(2)}$, we can rely on the formula for the Gauss-Seidel method:

$$x_i^{(k)} = \frac{1}{a_{i,i}} \left(b_i - \sum_{j=1}^{i-1} a_{i,j} x_j^{(k)} - \sum_{j=i+1}^n a_{i,j} x_j^{(k-1)} \right)$$

Let us start from $x^{(1)} = [x_1^{(1)}, x_2^{(1)}, x_3^{(1)}]$:

$$x_1^{(1)} = \frac{1}{a_{1,1}} \left(b_1 - \sum_{j=2}^3 a_{1,j} x_j^{(0)} \right) = \frac{1}{4} (0 - 2 \cdot 0 - 0 \cdot 0) = 0$$

$$x_2^{(1)} = \frac{1}{a_{2,2}} \left(b_2 - a_{2,1} x_1^{(1)} - a_{2,3} x_3^{(0)} \right) = \frac{1}{2} (2 - 0 \cdot 0 - 1 \cdot 0) = 1$$

$$x_3^{(1)} = \frac{1}{a_{3,3}} \left(b_3 - \sum_{j=1}^2 a_{3,j} x_j^{(1)} \right) = \frac{1}{4} (0 - 1 \cdot 0 - (-1) \cdot 1) = \frac{1}{4}$$

So $x^{(1)} = [0, 1, \frac{1}{4}]$. Now we can compute $x^{(2)} = [x_1^{(2)}, x_2^{(2)}, x_3^{(2)}]$:

$$x_1^{(2)} = \frac{1}{a_{1,1}} \left(b_1 - \sum_{j=2}^3 a_{1,j} x_j^{(1)} \right) = \frac{1}{4} \left(0 - 2 \cdot 1 - 0 \cdot \frac{1}{4} \right) = -\frac{1}{2}$$

$$x_2^{(2)} = \frac{1}{a_{2,2}} \left(b_2 - a_{2,1} x_1^{(2)} - a_{2,3} x_3^{(1)} \right) = \frac{1}{2} \left(2 - 0 \cdot \left(-\frac{1}{2} \right) - 1 \cdot \frac{1}{4} \right) = \frac{7}{8}$$

$$x_3^{(2)} = \frac{1}{a_{3,3}} \left(b_3 - \sum_{j=1}^2 a_{3,j} x_j^{(2)} \right) = \frac{1}{4} \left(0 - 1 \cdot \left(-\frac{1}{2} \right) - (-1) \cdot \frac{7}{8} \right) = \frac{11}{32}$$

So $x^{(2)} = [-\frac{1}{2}, \frac{7}{8}, \frac{11}{32}]$.

Exercise 3

Solve the linear system $Ax = b$ using the Conjugate Gradient method, with:

$$A = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solution

Since $A \in \mathbb{R}^{3 \times 3}$, we observe that, in exact arithmetic, Conjugate Gradient Method will converge in at most 3 iteration. Let us recall the pseudocode for the Conjugate Gradient Method:

$$r^{(0)} = b - Ax^{(0)}$$

$$p^{(0)} = r^{(0)}$$

for $k = 1, 2, \dots$, maxiter:

$$y = Ap^{(k-1)}$$

$$\alpha_{k-1} = (p^{(k-1)}, r^{(k-1)}) / (y, p^{(k-1)})$$

$$x^{(k)} = x^{(k-1)} + \alpha_{k-1}p^{(k-1)}$$

$$r^{(k)} = b - Ax^{(k)} = r^{(k-1)} - \alpha_{k-1}y$$

$$\beta_{k-1} = (y, r^{(k)}) / (y, p^{(k-1)})$$

$$p^{(k)} = r^{(k)} - \beta_{k-1}p^{(k-1)}$$

If Stopping criteria are satisfied exit the loop

end

Let us start by computing the initial residual $r^{(0)}$, that is also the initial descent direction $p^{(0)}$:

$$r^{(0)} = b - Ax^{(0)} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad p^{(0)} = r^{(0)} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Now, we need to compute the initial step length α_0 , to do that we firstly compute the quantity $Ap^{(0)}$:

$$Ap^{(0)} = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix}$$

At this point, we can compute α_0 :

$$\alpha_0 = \frac{(p^{(0)}, r^{(0)})}{(Ap^{(0)}, p^{(0)})} = \frac{\begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}{\begin{bmatrix} -2 & -4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}} = \frac{6}{4} = \frac{3}{2}$$

We are now able to compute the first approximate solution $x^{(1)}$ as

$$x^{(1)} = x^{(0)} + \alpha_0 \cdot p^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \frac{3}{2} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -3 \\ 3/2 \end{bmatrix}$$

and the next residual $r^{(1)}$ is

$$r^{(1)} = b - Ax^{(1)} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 3/2 \\ -3 \\ 3/2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} -3 \\ -6 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}$$

Now, to perform the next iteration, we need to determine the next descent direction $p^{(1)} = r^{(1)} - \beta_0 \cdot p^{(0)}$. Hence, we need to compute β_0 :

$$\beta_0 = \frac{(Ap^{(0)}, r^{(1)})}{(Ap^{(0)}, p^{(0)})} = \frac{\begin{bmatrix} -2 & -4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}}{\begin{bmatrix} -2 & -4 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}} = -\frac{32}{4} = -8$$

At this point, we obtain

$$p^{(1)} = r^{(1)} - \beta_0 \cdot p^{(0)} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} - (-8) \cdot \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 \\ -12 \\ 12 \end{bmatrix}$$

Now, we need to compute the next step length α_1 , to do that we firstly compute the quantity $Ap^{(1)}$:

$$Ap^{(1)} = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 12 \\ -12 \\ 12 \end{bmatrix} = \begin{bmatrix} 24 \\ 24 \\ 24 \end{bmatrix}$$

and thus α_1 is

$$\alpha_1 = \frac{(p^{(1)}, r^{(1)})}{(Ap^{(1)}, p^{(1)})} = \frac{\begin{bmatrix} 12 & -12 & 12 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}}{\begin{bmatrix} 24 & 24 & 24 \end{bmatrix} \cdot \begin{bmatrix} 12 \\ 12 \\ 12 \end{bmatrix}} = \frac{48}{24 \cdot 12 \cdot 3} = \frac{1}{6}$$

We are now able to compute the second approximate solution $x^{(2)}$ as

$$x^{(2)} = x^{(1)} + \alpha_1 \cdot p^{(1)} = \begin{bmatrix} 3/2 \\ -3 \\ 3/2 \end{bmatrix} + \frac{1}{6} \cdot \begin{bmatrix} 12 \\ -12 \\ 12 \end{bmatrix} = \begin{bmatrix} 7/2 \\ -5 \\ 7/2 \end{bmatrix}$$

and the next residual is

$$r^{(2)} = b - Ax^{(2)} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 4 \\ 1 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 7/2 \\ -5 \\ 7/2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since we have obtained the null vector as the residual, we have that $x^{(2)}$ is the exact solution.